

Restoration of waterways and natural drainage networks

Spatial scale: landscape

About this sustainable management strategy

Rainfall moves across vineyards, roads, forests, and agricultural land before reaching streams and rivers. Managing these flow pathways is essential for reducing erosion, limiting flood risks, and maintaining water availability across the landscape. Ecological waterways and vegetated runoff management restore natural drainage features so that water is slowed, filtered, and gradually absorbed instead of rapidly leaving the catchment.

Practical actions include restoring riparian vegetation, establishing grassed waterways and swales, reconnecting small floodplain areas, and maintaining vegetated buffer strips alongside streams and drainage channels. Existing runoff channels can be reshaped and stabilised with permanent vegetation to safely convey excess water during heavy rainfall while reducing flow velocity and sediment transport. Sediment retention areas and vegetated infiltration zones can be established where runoff naturally concentrates to capture eroded soil and promote water infiltration before water reaches streams or rivers. Traditional landscape features such as dry-stone walls also contribute by slowing runoff, improving hillside stability, and supporting natural drainage.

These measures are most effective when implemented across the wider catchment and combined with complementary soil management practices within vineyards, such as permanent ground cover, cover cropping, and reduced tillage. Together, they strengthen the landscape's capacity to regulate water, reduce the impacts of extreme rainfall, and improve resilience to longer dry periods.

Key ecosystem services:

- Flood and erosion control (*reduced runoff and sediment transport*)
- Water regulation (*enhanced infiltration, groundwater recharge and natural flow regulation*)
- Water quality improvement
- Drought mitigation through increased landscape water retention
- Biodiversity enhancement (*above- and below-ground habitats and ecological connectivity*)
- Cultural heritage (*maintenance of traditional landscape features*)
- Landscape aesthetics (*enhanced visual quality and landscape character*)

Considerations for successful implementation

Alignment with natural drainage patterns

Waterways, swales, and buffer zones should follow existing topography and natural flow paths. Preserving or restoring these features is generally more effective than replacing them with artificial drainage infrastructure.

Selection and management of vegetation

Native grasses, shrubs, and riparian plant species provide year-round soil protection, improve bank stability, and create habitat for wildlife. Vegetation should be selected according to local site conditions and managed to maintain both ecological function and water conveyance.

Landscape-scale coordination

Many benefits, including flood mitigation and improved water quality, depend on coordinated management across neighbouring properties and the wider catchment. Collaboration among landowners and local authorities can significantly increase the effectiveness of these measures.

Regular inspection and maintenance

Regular inspections help identify erosion, sediment build-up, blocked flow paths, or damaged vegetation before these reduce the performance of the system. Early maintenance usually requires less effort than repairing severe erosion or channel degradation.